Data Quality Over Model Complexity: Rethinking the AI Performance Paradigm

Author: Michael Adelusola

Date: December 2025

Abstract

The rapid advancement of artificial intelligence has fueled a global race toward building increasingly large and complex models, often under the assumption that scale alone guarantees superior performance. Yet across industries, AI systems continue to fail—not because models are too small, but because the data feeding them is noisy, incomplete, biased, or inconsistently labeled. This paper challenges the prevailing model-centric mindset by arguing that high-quality, well-curated data is a far more powerful driver of AI success than incremental increases in model size. Drawing on evidence from machine learning, data engineering, applied analytics, and real-world deployments, the paper demonstrates how clean and reliable data dramatically improves accuracy, robustness, interpretability, and downstream decision quality. It further examines how poor data hygiene undermines even state-of-the-art architectures, leading to hallucinations, instability, biased predictions, and operational failures. The paper advocates for a shift toward data-centric AI practices, highlighting emerging methods in automated data cleaning, anomaly detection, human-in-the-loop workflows, and governance frameworks that systematically elevate data quality. By rethinking the AI performance paradigm, this work underscores that the most transformative gains in AI do not come from building bigger models, but from building better data.

2. Introduction

Over the past decade, artificial intelligence has experienced explosive growth, driven largely by increasingly large and complex model architectures. Organizations across industries now invest heavily in scaling model parameters, expanding compute clusters, and training ever-deeper neural networks in pursuit of higher accuracy and competitive advantage. Yet, despite these escalating efforts, many AI systems continue to deliver inconsistent, unreliable, or even misleading outputs. This performance gap reveals an uncomfortable truth: most AI failures are not caused by model limitations but by poor-quality data. As Abbas and Hussain (2025) argue, clean, reliable data remains the real foundation of successful AI, even though it is frequently overshadowed by the pursuit of larger models.

Low-quality data introduces noise, bias, incompleteness, and inconsistencies that fundamentally distort the learning process. Models trained on incorrect, missing, or unrepresentative samples inevitably internalize flawed assumptions, leading to hallucinations, weak generalization, and brittle performance. This issue is amplified in relational datasets, where overlooked dependencies or missing attribute values undermine even sophisticated modeling pipelines (Zhu et al., 2025).

Likewise, research in applied data analytics shows that poor data hygiene compromises insights in business intelligence, educational analytics, and institutional decision-making (Esomonu, 2025; Pathak, 2025). Across domains such as healthcare, environmental modeling, manufacturing, and cybersecurity, studies consistently reveal that AI systems fail not because models are too shallow but because the data beneath them is incomplete or polluted (Fairfield, 2025; Ahangar et al., 2025; Machireddy, 2025; Chowdhury et al., 2025).

Despite this evidence, organizations often default to scaling models rather than addressing foundational data issues. Pagliaro (2025) notes that modern AI culture has normalized the belief that more data and larger architectures inherently improve performance, leading to a "bigger is better" mentality that overlooks critical data quality challenges. Yet, automated cleaning and AI-driven preprocessing pipelines have already demonstrated their ability to dramatically enhance dataset reliability and downstream model performance (Kilani et al., 2025). Similarly, advances in anomaly detection, automated annotation, and human-in-the-loop validation show that data refinement—rather than model expansion—is often the key differentiator between successful and failed AI deployments (Ahi et al., 2025; Karim et al., 2025).

The growing body of work around data-centric AI reinforces this shift, emphasizing that the most impactful performance gains often stem from improving data completeness, consistency, validity, and semantic clarity rather than increasing model size (Veluru et al., 2025). Industry research also highlights significant economic incentives for prioritizing data quality: organizations that effectively harness clean data outperform competitors, reduce operational waste, and deploy more trustworthy AI solutions (Islam, 2025). Even in complex scientific and engineering contexts—such as off-grid energy forecasting, climate modeling, and IoT network security—models consistently succeed when the underlying data is curated, structured, and validated (Rayhan, 2025; Sham et al., 2025; Sood et al., 2025).

This paper argues for a paradigm shift away from model-centric thinking and toward a data-first AI strategy. It synthesizes contemporary research, examines the shortcomings of model-scaling approaches, and proposes a structured framework for organizations seeking to maximize AI performance through systematic data quality improvement. By re-centering data as the primary determinant of AI success, this work challenges the dominant narrative within the AI community and encourages a more effective, sustainable, and scientifically grounded approach to building intelligent systems.

3. Background and Problem Context

The AI community has long celebrated breakthroughs driven by increasingly large model architectures, from deep neural networks to massive transformer-based systems. This model-centric paradigm emphasizes algorithmic sophistication, architectural scaling, and computational power as the primary levers for improving performance. However, as emerging research indicates, scaling alone cannot compensate for low-quality data. Without clean, consistent, and representative datasets, even the most advanced models will encode distorted patterns, propagate bias, and generate unreliable predictions (Abbas & Hussain, 2025).

3.1 Rise of Model-Centric AI Development

The push toward bigger models is partly cultural and partly commercial. Pagliaro (2025) argues that the AI ecosystem has normalized a belief in scale as a dominant performance driver—assuming that larger datasets, broader feature spaces, and deeper architectures inherently guarantee better predictions. This mindset aligns with high-profile successes in natural language processing and computer vision but overlooks a key reality: many business and scientific datasets do not follow the properties of those large benchmark corpora. As a result, scaling models in these environments often leads to diminishing returns or complete performance collapse when data irregularities dominate the learning process.

Furthermore, relational and structured datasets present unique challenges. Zhu et al. (2025) note that missing attributes, relational inconsistencies, and unresolved entity-linking problems severely undermine downstream learning, regardless of the model used. This illustrates a crucial limitation of model-centric strategies—they assume data completeness, accuracy, and consistency as default conditions, when in practice, most real-world systems violate these assumptions.

3.2 Challenges of Low-Quality Data in AI Systems

AI systems trained on poor-quality data routinely fail in ways that are subtle, harmful, and expensive to detect. In applied business analytics, Pathak (2025) highlights that organizations often underestimate the importance of data validity and consistency, resulting in misleading insights and inaccurate forecasts. In scientific domains, low-quality measurements lead to incorrect interpretations, model drift, and faulty predictions, which Fairfield (2025) characterizes as "invisible pollution" that quietly degrades AI reliability.

The consequences are especially severe in fields where data incompleteness or noise directly affects human outcomes. Healthcare studies show that missing or mislabeled clinical data causes predictive models to miss early indicators of disease or misclassify treatment pathways (Machireddy, 2025; Arzikulov & Komiljonov, 2025). Similarly, in industrial engineering contexts, poor-quality sensor data makes anomaly detection and forecasting models significantly less effective (Chowdhury et al., 2025). These failures reinforce the fundamental premise that AI accuracy is bounded more by data fidelity than by model complexity.

Kilani et al. (2025) add that increased data volume often amplifies existing problems rather than solving them. Scaling dirty datasets leads to larger, more complex errors, making models harder to interpret and harder to correct.

3.3 Why Bigger Models Cannot Fix Bad Data

The misconception that larger models can "learn around" flawed data persists, but empirical evidence consistently shows otherwise. Ongoing work in education analytics, energy systems, and IoT security underscores that models can only extract meaningful patterns if the underlying data reflects real-world behavior accurately (Esomonu, 2025; Rayhan, 2025; Sood et al., 2025). When data is inconsistent or sparse, increasing model size merely increases the likelihood of overfitting, hallucination, instability, and misclassification.

Moreover, operational AI environments introduce additional complexities that scaling cannot resolve:

- **Bias amplification** caused by unrepresentative or skewed datasets
- **Signal loss** due to noisy, redundant, or inaccurately annotated samples
- **Poor generalization** when rare but important features are overlooked (Ahangar et al., 2025)
- **Model fragility** when environmental factors shift slightly

As Veluru et al. (2025) argue, the central flaw in the model-centric approach is its false assumption that architecture alone drives AI performance. In reality, the quality of the underlying data—not its quantity or the model's size—sets the upper bound for what AI systems can achieve.

4. Data-Centric AI: Foundations and Principles

Data-centric AI represents a paradigm shift away from the traditional focus on expanding model complexity and toward systematically improving the quality, consistency, and relevance of the data powering those models. Unlike model-centric development—which assumes data is a static resource to be consumed—data-centric AI views data as the primary lever for improving AI performance. This approach argues that more reliable, structured, and representative datasets lead not only to higher accuracy but also to safer, more interpretable, and more trustworthy AI systems (Veluru et al., 2025).

4.1 What Is Data-Centric AI?

Data-centric AI emphasizes refining datasets rather than inflating model parameters. Veluru et al. (2025) describe it as a rebalancing of priorities: instead of optimizing architecture first, practitioners first optimize the data pipeline—cleaning, validating, labeling, and structuring data so that models have high-quality signals to learn from.

This shift acknowledges several realities:

- Noise and missing values degrade learning more than architectural limitations.
- Structured, curated datasets reduce overfitting and improve generalization.

- High-quality labels outperform added model layers in many supervised tasks.

In other words, data quality—not model size—determines the ceiling of AI performance.

4.2 Data Quality Dimensions

Pathak (2025) outlines several key attributes that define high-quality datasets:

- **Accuracy** – Data reflects real-world conditions faithfully.
- **Completeness** – Missing values, gaps, and absent attributes are minimized.
- **Consistency** – Formats, schemas, and semantics are uniform across sources.
- **Validity** – Data adheres to defined rules, units, and constraints.
- **Timeliness** – Up-to-date records reduce prediction errors.
- **Lineage and Trustworthiness** – Transparent data provenance supports governance.

These dimensions provide a systematic foundation for evaluating and improving the datasets that drive AI models.

4.3 Automated Data Cleaning Techniques

Recent research highlights significant advances in automated data cleaning driven by AI and machine learning. Kilani et al. (2025) report that ML-based cleaning methods—such as imputation, anomaly remediation, deduplication, entity resolution, and automated constraint inference—significantly improve data reliability, particularly for high-volume or complex systems.

AI-powered cleaning methods provide four major advantages:

1. **Scalability** – Capable of handling large datasets at speeds impossible for manual review.
2. **Adaptiveness** – Models learn domain-specific cleaning rules over time.
3. **Consistency** – Automated processes reduce human error and subjective judgment.
4. **Precision** – Deep-learning methods can detect subtle relational inconsistencies and missing patterns.

These findings support the growing consensus that robust cleaning pipelines often yield higher performance improvements than adding extra model layers.

4.4 Human-in-the-Loop Data Validation

While automated tools enhance scalability and precision, human expertise remains essential for resolving ambiguous or context-sensitive issues. Ahi et al. (2025) demonstrate how human-in-the-loop (HITL) systems dramatically boost data quality—achieving over 80% gains in reviewer productivity when assisted by AI-driven anomaly detection and refinement tools. These HITL pipelines allow humans to validate nuanced cases, correct model mistakes, and enforce domain-specific guidelines that automated systems might overlook.

Similarly, Karim et al. (2025) emphasize that AI agents can accelerate annotation, reasoning, and data refinement, but hybrid human–AI systems remain the most effective solution for ensuring semantic accuracy.

Together, automated cleaning and supervised validation form the backbone of robust data-centric AI practices.

6. Proposed Framework: Data-First Pipeline for Reliable AI Performance

The empirical evidence presented across multiple domains demonstrates that AI systems perform best when built on clean, well-curated, and contextually meaningful data. To operationalize this insight, this section introduces a **Data-First AI Pipeline**—a structured framework that prioritizes data quality at every stage of the machine learning lifecycle. The framework incorporates automated validation techniques, intelligent cleaning mechanisms, human oversight, and continuous monitoring to ensure that data remains trustworthy, consistent, and aligned with real-world conditions.

6.1 Data Validation Layer

The first layer of the framework focuses on detecting missing values, inconsistencies, structural violations, and anomalies within raw datasets. Automated validation techniques identify:

- incomplete attributes,
- duplicate records,
- out-of-range values,
- structural misalignments (e.g., schema drift),
- logical inconsistencies.

These problems are prevalent across domains and frequently go undetected in traditional pipelines, as highlighted in relational data cleaning research (Zhu et al., 2025). Sood et al. (2025) further demonstrate that missing data is especially harmful in IoT and cybersecurity contexts, where even small gaps can compromise anomaly detection and degrade robustness. By integrating validation early, organizations prevent flawed data from propagating downstream into training and inference phases.

6.2 Automated Anomaly Detection and Cleaning

Once validation surfaces inconsistencies, the next step is automated cleaning powered by AI and machine learning. Kilani et al. (2025) outline several ML-driven cleaning strategies—imputation, deduplication, constraint inference, semantic normalization, and relational consistency enforcement—that dramatically enhance data reliability.

Key advantages of automated cleaning in this framework include:

- **Speed and scalability:** Automation handles large datasets impossible to review manually.
- **Precision:** ML models detect subtle irregularities, such as relational mismatches or soft anomalies.
- **Domain adaptiveness:** Over time, models learn how to clean data within specific domains more effectively.

These automated approaches align with industrial findings showing that poor-quality data causes failures that larger models cannot compensate for (Chowdhury et al., 2025; Rayhan, 2025).

6.3 Continuous Feedback Loop for Data Integrity

A robust AI system cannot depend on a one-time cleaning process. New patterns, environmental changes, and domain-specific drift require continuous monitoring. Esomonu (2025) stresses that ongoing data assurance practices are essential, especially in dynamic institutional or organizational environments where data quality affects long-term outcomes.

The continuous feedback loop in this framework includes:

- **Drift detection:** Identifying when new data deviates from historical norms.
- **Adaptive cleaning:** Automatically updating cleaning rules as distributions shift.
- **Performance-based monitoring:** Using model errors as indicators of upstream data issues.
- **Periodic retraining:** Ensuring models are aligned with current, not historical, data landscapes.

Such iterative refinement prevents gradual model degradation, a common problem when organizations focus solely on model expansion.

6.4 Governance and Documentation Requirements

To ensure transparency, accountability, and reproducibility, the framework incorporates strong data governance principles. Islam (2025) notes that organizations that systematically harness and document their data pipelines outperform those that treat data as an afterthought. Likewise, Xu et al. (2025) highlight that trustworthy AI systems must maintain verifiable data lineage, consistent metadata standards, and fully documented transformations.

This governance layer includes:

- **Version-controlled data repositories**,
- **Lineage tracking**,
- **Documentation of cleaning rules**,
- **Validation logs**,
- **Clear definitions of quality thresholds**,
- **Policy enforcement mechanisms**.

Together, these elements strengthen compliance, auditability, and long-term integrity—critical for AI deployed in regulated industries like healthcare, finance, or manufacturing.

7. Discussion

The findings and framework presented in this paper highlight a fundamental shift needed within the AI ecosystem: organizations must rethink their reliance on model-centric development and prioritize systematic improvements in data quality. Despite overwhelming evidence that clean, reliable data produces stronger, more trustworthy AI outcomes, many institutions remain fixated on scaling models, adopting larger architectures, or purchasing additional compute resources. This section examines why the misconception persists, identifies the true bottlenecks limiting AI performance, and outlines practical strategies for operationalizing a data-centric mindset.

7.1 Why Organizations Keep Chasing Bigger Models

The global AI landscape has glamorized large-scale model architectures, positioning them as symbols of technological superiority and competitive advantage. Pagliaro (2025) attributes this trend to the success of models in big data domains like NLP and vision, which has created a misconception that scale universally translates to better performance. This illusion persists because organizations often overlook the hidden costs of dirty data. As Abbas and Hussain (2025) describe, many failures attributed to "model limitations" are actually rooted in upstream data noise, missing values, or structural inconsistencies.

Additionally, organizations may prioritize model expansion because:

- **Computational investments are more visible** than data quality improvements.
- **Data issues are harder to diagnose**, especially when buried in complex pipelines.
- **Vendors and research narratives overemphasize model scaling** rather than provenance and curation.
- Teams often **assume data is “good enough”**, even when evidence suggests otherwise (Zhu et al., 2025).

This creates a cycle in which flawed models are continuously retrained, expanded, or refactored, while the underlying data issues remain unresolved.

7.2 The Real Technical Bottleneck: Data, Not Models

Studies across engineering, healthcare, cybersecurity, and scientific modeling repeatedly show that data—not models—is the primary determinant of AI performance. In healthcare, low-quality clinical data leads directly to missed diagnoses or inaccurate predictions, regardless of model size (Machireddy, 2025; Arzikulov & Komiljonov, 2025). In manufacturing and anomaly detection contexts, incomplete datasets cause AI systems to overlook rare but critical events (Ahangar et al., 2025). Even environmental and climate models collapse when trained on inconsistent ecological datasets (Fairfield, 2025; Sham et al., 2025).

Kilani et al. (2025) further illustrate that automated cleaning pipelines significantly outperform attempts to compensate for noise by increasing model depth. Moreover, incomplete or sparse datasets—common in IoT and cybersecurity—cause instability that scaling cannot fix (Sood et al., 2025). These findings confirm that data quality sets the upper bound for performance; model complexity only fine-tunes the final curve.

7.3 Strategies to Shift to a Data-Centric Mindset

To move beyond a model-centric culture, organizations must adopt structural, cultural, and technological changes:

7.3.1 Leadership and Organizational Culture

Executives must recognize that data quality is not a “backend detail” but a primary AI investment. Islam (2025) shows that organizations leveraging data governance systematically achieve better business outcomes than those that prioritize only model engineering.

7.3.2 Tooling and Automation

Integrating data validators, anomaly detectors, and automated cleaning tools into pipelines ensures high-quality input at scale. Kilani et al. (2025) demonstrate that these systems outperform manual cleaning and lower operational costs.

7.3.3 Human-in-the-Loop Oversight

Ahi et al. (2025) and Karim et al. (2025) highlight that hybrid workflows significantly enhance accuracy and accelerate annotation by letting experts resolve edge cases AI cannot yet interpret.

7.3.4 Continuous Monitoring and Drift Detection

Esomonu (2025) emphasizes that maintaining data quality requires ongoing surveillance, not a one-time cleaning event. Drift detection, lineage tracking, and periodic audits must become standard practice.

7.3.5 Redesigning Evaluation Metrics

Model performance should be evaluated not only by accuracy but also by:

- input quality scores,
- data completeness indices,
- metadata consistency,
- annotation reliability metrics.

This reframes success around data stewardship rather than model scale.

8. Conclusion and Future Directions

This paper has argued that the dominant model-centric narrative in artificial intelligence overlooks a far more fundamental driver of performance: data quality. While organizations continue to invest in larger architectures, deeper networks, and increased computational capacity, the evidence is overwhelmingly clear that clean, consistent, and reliable data has a far greater impact on accuracy, robustness, interpretability, and long-term system success. As Abbas and Hussain (2025) emphasize, data—not model complexity—is the true foundation of AI reliability. Across scientific, industrial, business, and governmental domains, the failures and instabilities observed in AI deployments repeatedly stem from upstream data issues rather than algorithmic limitations (Fairfield, 2025; Pathak, 2025; Machireddy, 2025; Chowdhury et al., 2025).

By reviewing contemporary research, this paper has shown that relational inconsistencies (Zhu et al., 2025), incomplete or sparse measurements (Sood et al., 2025), poor annotations (Karim et al., 2025), and systematic noise (Sham et al., 2025) undermine even the most advanced architectures. Automated cleaning pipelines (Kilani et al., 2025), hybrid human–AI validation workflows (Ahi et al., 2025), and well-governed data practices (Islam, 2025) produce far more meaningful improvements than simply scaling models. The proposed **Data-First AI Pipeline** integrates validation, cleaning, feedback loops, and governance principles to offer a practical blueprint for organizations seeking to redirect their AI strategy toward data-centric excellence.

8.1 Future Research Directions

While the shift toward data-centric AI is gaining momentum, several important research opportunities remain:

8.1.1 Autonomous Data Cleaning and Integrity Agents

Future systems may employ autonomous AI agents capable of detecting, repairing, and enriching data streams in real time. Karim et al. (2025) illustrate early progress in AI-driven annotation, hinting at the potential for fully automated, context-aware data governance.

8.1.2 Explainable Data Quality Metrics

There is a growing need for interpretable data quality scores that quantify the reliability of training and inference datasets in a standardized way. Such metrics would support regulatory frameworks, model auditing, and organizational transparency.

8.1.3 Domain-Specific Data Quality Frameworks

Sectors like healthcare, climate modeling, and cybersecurity require specialized cleaning and validation approaches tailored to their data ecosystems (Arzikulov & Komiljonov, 2025; Xu et al., 2025; Rayhan, 2025). Developing domain-targeted frameworks will be critical for achieving safe and reliable AI outcomes.

8.1.4 Integrating Data-Centric and Model-Centric Paradigms

While this paper argues strongly for data-first methods, future research should explore hybrid strategies that integrate robust data curation with scalable architectures to maximize both efficiency and performance.

8.1.5 Ethical and Governance Implications of Data Quality

As AI systems become more embedded in society, poor data stewardship can lead to discrimination, exclusion, and systemic harm. Future work must examine the ethical, social, and policy dimensions of data-centric AI to ensure equitable outcomes.

8.2 Final Perspective

The next wave of AI progress will not be defined by bigger models, but by **better data**. Clean, reliable, well-governed datasets amplify model performance, reduce operational risks, enhance trustworthiness, and ensure more stable deployment across real-world conditions. By embracing data-centric AI, organizations can build systems that are not only more accurate, but also safer, fairer, and more aligned with human and institutional needs. As the evidence throughout this paper shows, high-quality data is not simply a technical requirement—it is the cornerstone of sustainable AI success.

References

1. Syed Abbas, Ali Azghar Hussain. (2025). The Overlooked Key to AI Success: Why Clean, Reliable Data Outperforms Bigger Models. International Journal For Multidisciplinary Research. 7. 10.36948/ijfmr.2025.v07i06.61533.
2. Ahangar, M. N., Farhat, Z. A., & Sivanathan, A. (2025). AI trustworthiness in manufacturing: Challenges, toolkits, and the path to Industry 5.0. *Sensors*. https://www.mdpi.com/
3. Ahi, K., Wu, S., Fenger, G., & Mansour, S. (2025). Human-in-the-loop system boosting reviewer productivity by 80%+ and accelerating decision-making via real-time anomaly detection and data refinement. *Metrology and Inspection*. https://spiedigitallibrary.org/
4. Arzikulov, F., & Komiljonov, A. (2025). *The role of artificial intelligence in personalized oncology: Predictive models and treatment optimization*. Academic Journal of Science. integrumpublication.org
5. Chowdhury, S., Ghosh, A., Acharya, S., & others. (2025). Transforming chemical process engineering: The role of AI and machine learning in revolutionizing process systems. *Canadian Journal of Chemical Engineering*. Wiley.
6. Esomonu, N. P. M. (2025). *Utilizing AI and big data for predictive insights on institutional performance and student success: A data-driven approach to quality assurance*. In *AI and Ethics, Academic Integrity and the Future of Education*. ResearchGate.
7. Fairfield, J. (2025). *Clean data: Recursion as pollution in environmental AI*. Oxford Intersections: AI in Society. scholarlycommons.law.wlu.edu
8. Islam, M. R. (2025). *AI and analytics for entrepreneurs: A practical guide to smarter business growth*. Google Books.
9. Karim, M. M., Khan, S., Van, D. H., Liu, X., Wang, C., & Qu, Q. (2025). Transforming data annotation with AI agents: A review of architectures, reasoning, applications, and impact. *Future Internet*. MDPI.
10. Kilani, S. O., Amao, I. K., Ojo, N. A., & Samson, P. A. (2025). *AI and machine learning-powered automated data cleaning methods: Improving data quality*. ijarpr.com
11. Machireddy, J. (2025). *Harnessing AI and data analytics for smarter healthcare solutions*. SSRN.
12. Pagliaro, A. (2025). Artificial intelligence vs. efficient markets: A critical reassessment of predictive models in the big data era. *Electronics*. MDPI.

13. Pathak, A. (2025). *Leveraging AI for better data quality and insights*. Journal of Computer Science and Technology Studies. al-kindipublishers.org
14. Rayhan, F. (2025). AI-enabled energy forecasting and fault detection in off-grid solar networks for rural electrification. *Authorea Preprints*. techrxiv.org
15. Sham, F. A. F., El-Shafie, A., Jaafar, W. Z. W., & Sherif, M. (2025). Advances in AI-based rainfall forecasting: A comprehensive review of past, present, and future directions with intelligent data fusion and climate change models. *Results in Engineering*. Elsevier.
16. Sood, K., Liu, S., Nguyen, D. D. N., & Kumar, N. (2025). Alleviating data sparsity to enhance AI models' robustness in IoT network security context. *IEEE Transactions on Network Science and Engineering*.
17. Sourav, M. S. A., Asha, N. B., & Reza, J. (2025). Generative AI in business analytics: Opportunities and risks for national economic growth. *Journal of Computer Science and Technology Studies*. al-kindipublishers.org
18. Veluru, S. R., Erukude, S. T., & Marella, V. C. (2025). *Data-centric AI: A systematic review of methods, challenges, and future directions*. International Journal of … papers.ssrn.com
19. Xu, D., Gondal, I., Yi, X., Susnjak, T., Watters, P., & others. (2025). The erosion of cybersecurity zero-trust principles through generative AI: A survey on challenges and future directions. *Journal of Cybersecurity and Privacy*. MDPI.
20. Zhu, J., Zhao, X., Sun, Y., Song, S., & others. (2025). Relational data cleaning meets artificial intelligence: A survey. *Data Science & Engineering*. EBSCOhost.